

# Customer Shopping Behavior Analysis Report

End-to-End Data Analytics Project

Python · PostgreSQL · Power BI

Prepared by: Data Analytics Team | June 2026

3,900  
Customers

18  
Columns

10  
SQL Queries

1  
Dashboard

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# 1. Business Problem Statement

A leading retail company wants to better understand its customers' shopping behavior in order to improve sales, customer satisfaction, and long-term loyalty. The management team has noticed changes in purchasing patterns across demographics, product categories, and sales channels. They are particularly interested in uncovering which factors — such as discounts, reviews, seasons, or payment preferences — drive consumer decisions and repeat purchases.

*"How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies?"*

# 2. Project Overview

This project follows a complete end-to-end data analytics workflow — from raw data ingestion to an interactive business dashboard — mirroring the standard methodology used in corporate analytics teams.

- 1. Data Preparation & Modeling (Python)** — Clean and transform the raw dataset for analysis using pandas.
- 2. Data Analysis (SQL)** — Load data into PostgreSQL, run structured queries to extract insights on customer segments, loyalty, and purchase drivers.
- 3. Visualization & Insights (Power BI)** — Build an interactive dashboard highlighting key patterns and trends for stakeholder decision-making.
- 4. Report & Presentation** — Summarize findings and business recommendations; prepare a stakeholder presentation deck.
- 5. GitHub Repository** — Host all Python scripts, SQL queries, and dashboard files in a well-structured public repository.

# 3. Dataset Summary

The dataset contains transactional and behavioral data for 3,900 customers across a range of product categories. Each row represents one customer and their most recent purchase interaction.

| Metric                 | Value              |
|------------------------|--------------------|
| Total Customers        | 3,900              |
| Dataset Columns        | 18                 |
| Product Categories     | Multiple           |
| Missing Values Handled | 37 (Review Rating) |

| Metric               | Value    |
|----------------------|----------|
| SQL Business Queries | 10       |
| Dashboard Tool       | Power BI |

## Key Feature Groups

- **Customer Demographics:** Age, Gender, Location, Subscription Status
- **Purchase Details:** Item Purchased, Category, Purchase Amount, Season, Size, Color
- **Shopping Behavior:** Discount Applied, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type, Payment Method

Missing Data: 37 null values were found exclusively in the Review Rating column and were handled using category-wise median imputation (see Section 4).

## 4. Exploratory Data Analysis — Python

All data preparation and initial exploration was performed in a Jupyter Notebook using the pandas library before loading the cleaned dataset into PostgreSQL.

### Data Loading

Imported the dataset using `pd.read_csv()` and reviewed the top rows with `df.head()`.

### Initial Exploration

Used `df.info()` to inspect data types and row counts, and `df.describe(include='all')` to obtain summary statistics for both numerical and categorical columns.

### Missing Value Treatment

Identified 37 null values in the `review_rating` column. Rather than applying an overall median — which could introduce bias across product categories — missing ratings were imputed using the median review rating within each product category. This preserves the natural distribution differences between categories.

### Column Standardization

All column names were converted to `snake_case` (lowercase with underscores) for consistency and compatibility with SQL queries.

### Feature Engineering

Two new columns were created: `age_group` (customers binned into Young Adult, Adult, Middle-Aged, Senior using `pd.qcut`) and `purchase_frequency_days` (textual frequency labels mapped to numeric day intervals for quantitative analysis).

### Redundancy Check

Verified that `discount_applied` and `promo_code_used` carried identical information for every row. The redundant `promo_code_used` column was dropped.

### Database Integration

Connected the Jupyter environment to PostgreSQL using SQLAlchemy and loaded the cleaned DataFrame into a customer table within the `customer_behavior` database.

## 5. Data Analysis — SQL

Ten structured business queries were executed in PostgreSQL to answer key analytical questions. Advanced SQL features including subqueries, CTEs, CASE statements, window functions (`ROW_NUMBER`), and aggregate functions were employed.

| # | Business Question         | Key Finding                                    |
|---|---------------------------|------------------------------------------------|
| 1 | Revenue by Gender         | Compared male vs. female total spend           |
| 2 | High-Spend Discount Users | Customers spending above avg despite discounts |

| #  | Business Question               | Key Finding                                        |
|----|---------------------------------|----------------------------------------------------|
| 3  | Top 5 Products by Rating        | Gloves, Sandals, Boots, Hat, Skirt                 |
| 4  | Standard vs. Express Shipping   | Express customers show higher avg. purchase        |
| 5  | Subscribers vs. Non-Subscribers | Similar avg. spend; non-subs drive more revenue    |
| 6  | Discount-Dependent Products     | 5 products with highest discount rate identified   |
| 7  | Customer Segmentation           | New / Returning / Loyal based on purchase history  |
| 8  | Top 3 per Category              | Best-sellers ranked by category using ROW_NUMBER() |
| 9  | Repeat Buyers & Subscriptions   | 2,500 repeat buyers still not subscribed           |
| 10 | Revenue by Age Group            | Young Adults drive the highest total revenue       |

## Notable Analytical Techniques

- Subquery to filter customers spending above the dataset average despite using discounts.
- CASE + SUM pattern to calculate percentage-based discount rates per product.
- CTE + CASE for multi-tier customer segmentation (New / Returning / Loyal).
- ROW\_NUMBER() window function partitioned by category to rank top products without ties.
- Subscription analysis filtered to repeat buyers (previous\_purchases > 5) to assess loyalty program effectiveness.

## 6. Dashboard — Power BI

An interactive Power BI dashboard was built directly on top of the PostgreSQL data source to provide stakeholders with a real-time, filterable view of customer behavior metrics.

### KPI Cards

- Number of Customers
- Average Purchase Amount (USD)
- Average Review Rating

### Charts & Visuals

- **Donut Chart:** Percentage of customers by subscription status.
- **Clustered Column Charts:** Revenue by Category and Sales by Category.
- **Clustered Bar Charts:** Revenue by Age Group and Sales by Age Group.

### Interactive Slicers

- Subscription Status
- Gender
- Product Category
- Shipping Type

All visuals respond dynamically to slicer selections, allowing stakeholders to drill down into specific customer segments and instantly observe the impact on revenue and sales KPIs.

## 7. Business Recommendations

Based on the combined findings from the SQL analysis and dashboard insights, the following strategic recommendations are proposed:

| Recommendation            | Rationale                                                                            |
|---------------------------|--------------------------------------------------------------------------------------|
| Boost Subscriptions       | Promote exclusive subscriber benefits to convert the large non-subscribed base.      |
| Customer Loyalty Programs | Reward repeat buyers to move them into the Loyal segment.                            |
| Review Discount Policy    | Some products rely heavily on discounts; balance sales boosts with margin control.   |
| Product Positioning       | Highlight top-rated and best-selling products (Gloves, Boots, Sandals) in campaigns. |
| Targeted Marketing        | Focus on high-revenue age groups (Young Adults) and Express-shipping users.          |

## 8. Conclusion

This project successfully demonstrated a full corporate-grade data analytics workflow applied to customer shopping behavior data. Key outcomes include:

- A clean, well-structured dataset ready for further analysis or machine learning applications.
- Ten SQL-driven business insights covering revenue, segmentation, product performance, and subscription behavior.
- An interactive Power BI dashboard enabling self-service exploration for business stakeholders.
- Actionable recommendations targeting subscription growth, loyalty programs, discount optimization, and demographic-focused marketing.

The methodology and code artifacts are available in the project's GitHub repository for full reproducibility and knowledge transfer.